DATA CASE STUDY

Type 2 diabetes awareness questionnaire

Analysis of the visitor journey, acquisition and outcomes

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Data Analyst / Data Engineer internship application · October 2026

Synthetic data · analytical categories, not medical diagnoses

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PLAN

What we will cover

01 Context & objectives of the analysis

02 Key issues & main findings

03 Data, quality & methodology

04 Key results of the analysis

05 Priority levers & recommendations

06 Open questions & outlook

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01

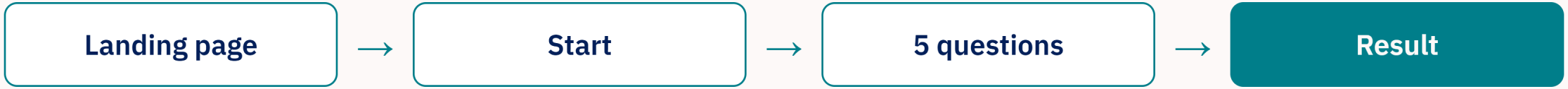
Context & objectives of the analysis

01 · CONTEXT

Doctoome's type 2 diabetes awareness page

- A dedicated page with a 5-question questionnaire on type 2 diabetes
- Visitors arrive through 8 channels: search, ads, social media, email, AI assistants...

VISITOR JOURNEY



3 POSSIBLE OUTCOMES

No current indication | **Possible risk** | Declared diagnosis

Scope: 26,000 sessions · 20,000 visitors · 1 June – 2 August 2026 · synthetic data

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01 · OBJECTIVES

Understanding how the page performs, and how to improve it

DOCTOOME'S GOAL

Identify people at possible risk and guide them towards a consultation.

OBJECTIVES OF THIS ANALYSIS

01 Where do we lose visitors along the journey?

02 Which channels bring the right profiles?

03 Can we trust the data?

04 What should Doctoome do next?

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02

Key issues & main findings

02 · KEY ISSUES

More than half of sessions never reach a result

43.7%

of sessions reach a result: **14,643 sessions** end along the way

35%

of sessions end **without even starting** the questionnaire

2.4×

gap between the best and worst campaign in possible-risk outcomes per session, **at equal age** (3.5× crude)

Every session lost represents a missed opportunity to engage users and generate a completed questionnaire.

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02 · MAIN FINDINGS

Four key findings and what they mean for Doctoome

35%

of sessions end before starting the questionnaire

→ Optimise the landing page

2×

More Q4 drop-offs on mobile

→ Investigate Q4 on mobile

13.0

possible-risk outcomes per 100 sessions with 45plus, at equal age (14.8 crude)

→ Test prioritising 45plus

+46%

possible-risk outcomes per session in 8 weeks, 57% from campaign mix

→ Validate through experimentation

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03

Data, quality & methodology

03 · DATA & METHODOLOGY

A four-step approach on 26,000 sessions

20,000
visitors

26,000
sessions

11,357
completed questionnaires

6
linked Parquet tables

METHODOLOGY

01
Audit
14 quality checks, nothing modified

02
Clean
SQL script, 4 clean tables, every fix verified

03
Analyse
Funnel, acquisition, outcomes, trends

04
Recommend
Prioritised, quantified levers

Python · DuckDB (SQL on Parquet) · pandas · matplotlib · fully reproducible notebook

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03 · DATA QUALITY

Reliable data after cleaning, but limited to 2 August

626

Sessions affected by incomplete data on 1–2 August

→ Trends on complete weeks only

3,531

Inconsistent categorical labels

→ Harmonised

697

Answers updated within ~10 seconds

→ Latest answer retained

324

Atypical sessions

→ Retained and flagged

No row deleted: every atypical case is flagged. Checked with no issue: unique IDs, links between tables, chronology, answer validity.

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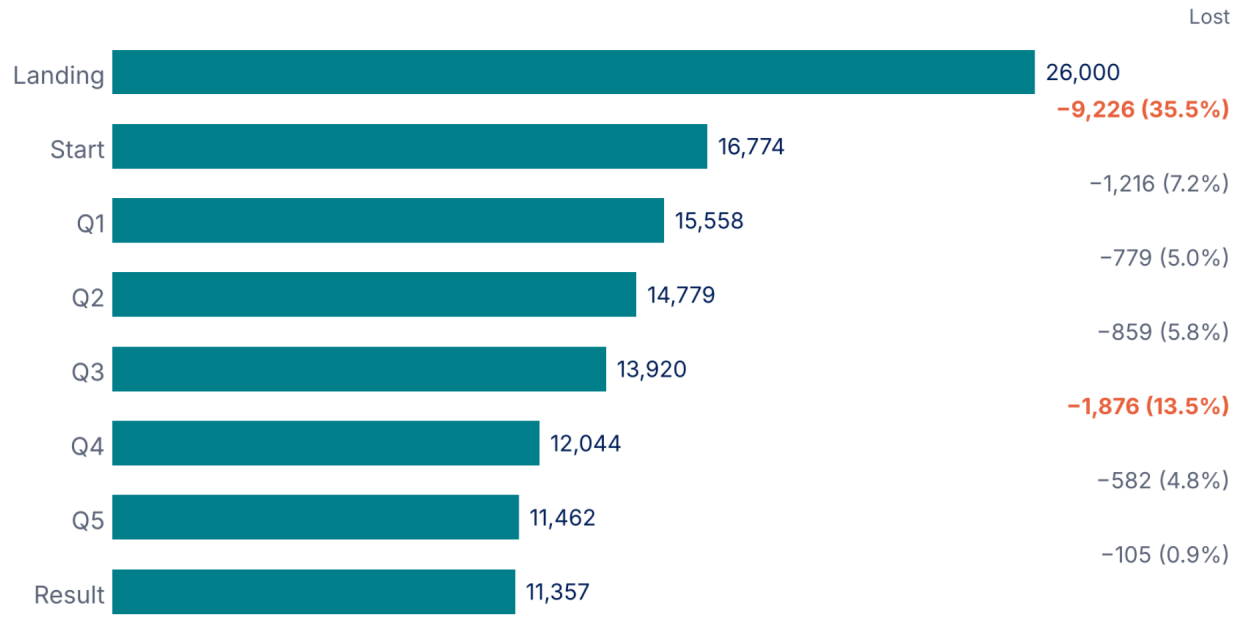
04

Key results of the analysis

04 · JOURNEY

Over a third of sessions end before starting

Sessions reaching each step, and sessions lost since the previous step



35.5%

of sessions end before starting: **63% of all losses**

43.7%

of sessions reach a result; **47.3% of unique visitors**

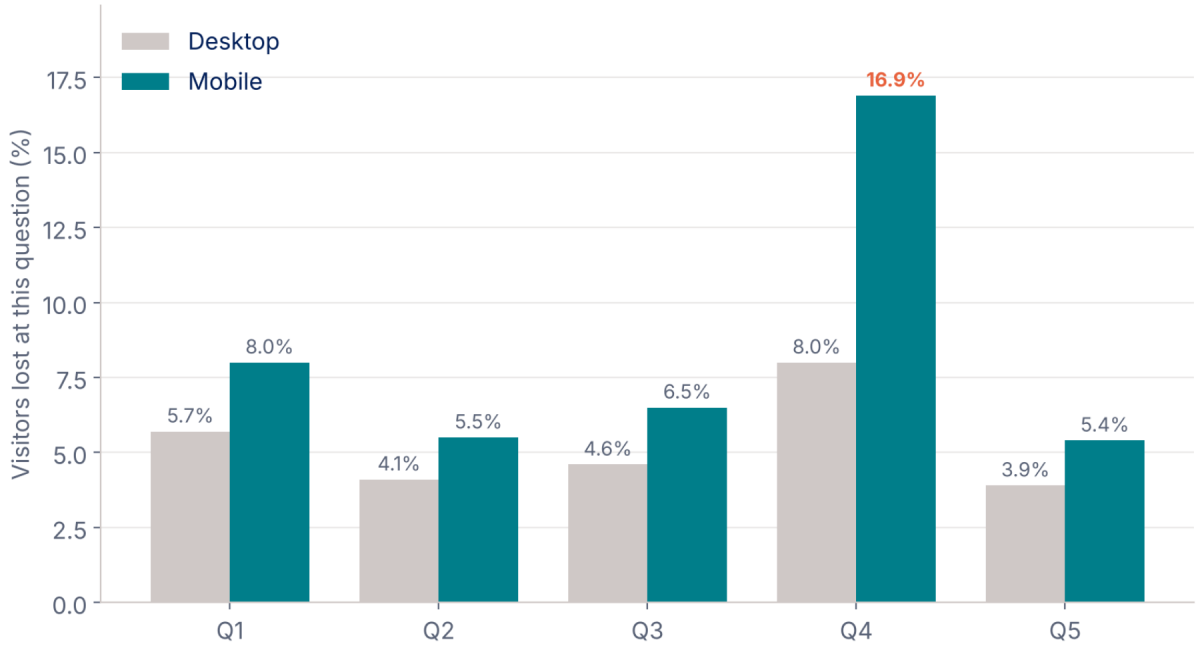
2nd leak: Q4. 13.5% of those who reach it stop there (vs ~5% elsewhere)

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04 · QUESTION 4

Q4 drop-off is twice as high on mobile

% of sessions that stop at each question, desktop vs mobile



16.9%

Q4 drop-off on mobile, vs **8.0% on desktop**

2.1×

the desktop rate, on a device that carries **65% of traffic**

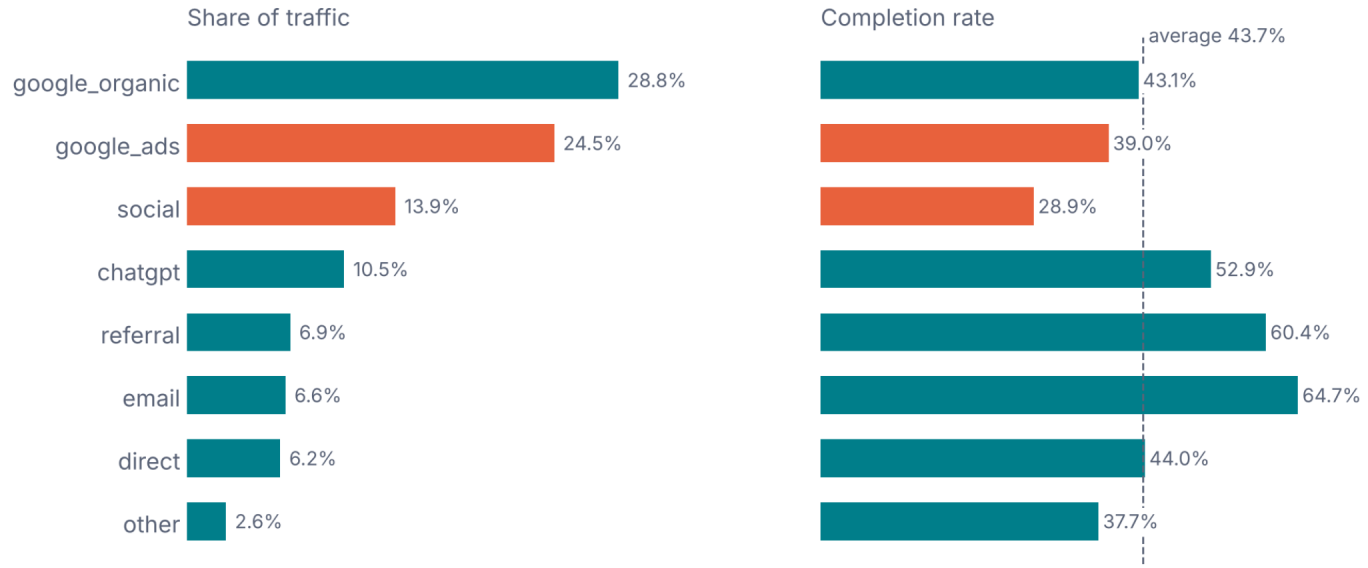
Not explained by the audience: the gap holds at equal age, source and returning mix. Cause still a hypothesis → UX review, then A/B test

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04 · ACQUISITION

Paid channels drive volume, but completion lags

Share of traffic and completion rate by source (orange: paid sources)



28.9%

completion for social, the weakest source (average 43.7%)

39.0%

for google_ads, the 2nd largest source. Best: email 64.7% (56.8% on new sessions), referral 60.4%

→ Compare channels on completion, volume and acquisition cost before reallocating budget

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04 · OUTCOMES

1 in 5 completers falls into the possible-risk category

20%

possible_risk: 2,276 results (2,190 people): a questionnaire category, not a diagnosis

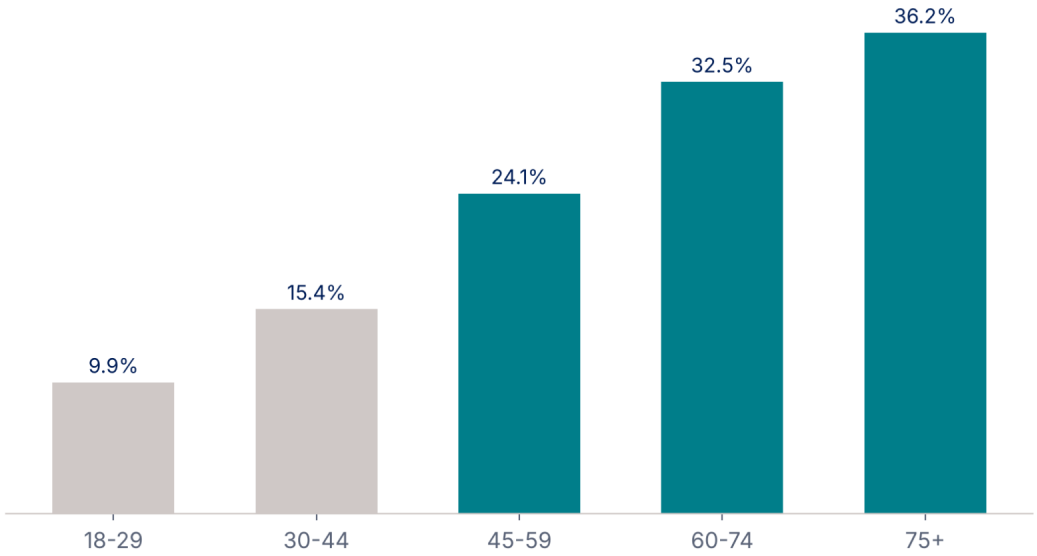
11%

declared_diagnosed: a follow-up need

69%

no_current_indication

Share of possible_risk within each age group (completed questionnaires): it rises with age

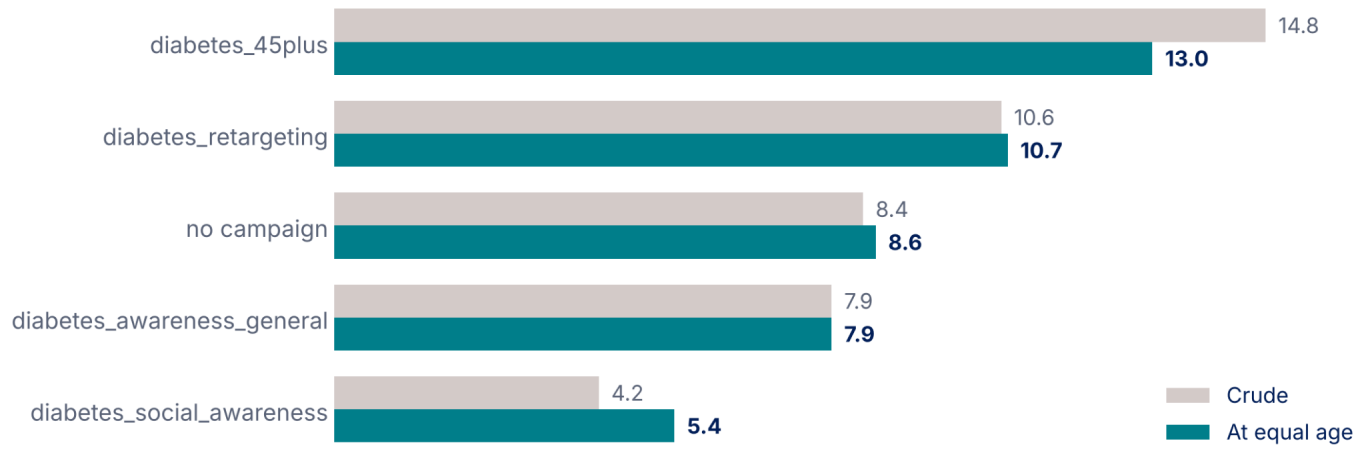


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04 · CAMPAIGNS

45plus leads, still 2.4× at equal age

possible-risk outcomes per 100 sessions, crude vs adjusted to the same age mix



13.0

for 45plus at equal age (14.8 crude): about **30% of its lead** comes from its older audience

5.4

for social_awareness at equal age (4.2 crude): young audience, low completion

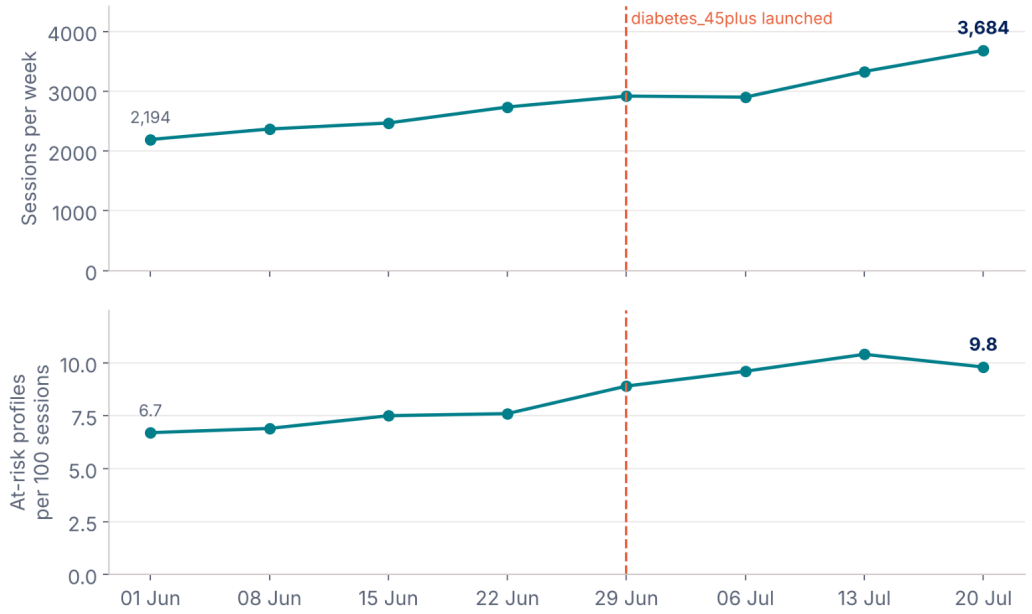
→ Test prioritising 45plus; assess cost and incremental impact first

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04 · TRENDS

Traffic and possible-risk outcomes per session both increased

Sessions per week (top) and possible_risk profiles per 100 sessions (bottom)



+68%

traffic (2,194 → 3,684 sessions), first vs last complete week

+46%

possible-risk outcomes per 100 sessions (6.7 → 9.8)

57% of the rise comes from the campaign mix, 43% from campaigns performing better → validate with a controlled test

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04 · ROBUSTNESS CHECKS

The findings hold once we control for the audience, but some gaps shrink

Q4 ON MOBILE

17.1% vs 8.0%

mobile vs desktop drop-off at the **same age, source and returning mix**: the gap is not explained by the audience

45PLUS CAMPAIGN

3.5× → 2.4×

lead over social_awareness **at equal age** (14.8 → 13.0 per 100 sessions)

EMAIL

64.7% → 56.8%

completion on **new sessions only**: 77% of email sessions are returning. Still among the best

WEEKLY TREND

57%

of the +46% rise comes from the **campaign mix**; 43% from campaigns performing better

Associations, not causes: these checks rule out the confounders we can observe. Each lever still needs a controlled test.

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05

Priority levers & recommendations

05 · RECOMMENDATIONS

Three priority experiments, one per team

01

Product

A/B test a clearer landing page to reduce the 35% pre-start drop-off

Investigate and test improvements to Q4 on mobile

Potential: up to ~700 additional results if mobile Q4 drop-off reached the desktop rate (scenario, upper bound)

02

Marketing

Test reallocating part of the budget from social_awareness to 45plus

Evaluate channels on completion, cost and possible-risk outcomes per 100 visits

Potential: 45plus keeps a lead at equal age (13.0 vs 5.4); confirm with costs and a controlled split

03

Data

Track consultation bookings after questionnaire completion

Collect campaign costs and set alerts for missing or inconsistent tracking

Potential: measure downstream conversion and cost per outcome, beyond completion alone

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06

Open questions & outlook

06 · OPEN QUESTIONS

What we still don't know, and the data to answer it

01

LANDING PAGE

Why do 35% leave before starting?

Collect: page load time, scroll depth, page version and entry source

02

MOBILE EXPERIENCE

What drives the higher Q4 drop-off on mobile?

Collect: screen size, time on Q4, interaction events and usability-test feedback

03

CAMPAIGN EFFECTIVENESS

Does 45plus deliver incremental outcomes at an acceptable cost?

Collect: campaign spend, complete UTMs and a suitable control group or experiment

04

DOWNSTREAM CONVERSION

Do users who receive a possible-risk outcome book a consultation?

Collect: booking events linked to the questionnaire session, with appropriate privacy safeguards

Priority: link questionnaire outcomes to consultation bookings

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06 · 12-MONTH OUTLOOK

One north-star metric, four supporting indicators

8.8

PROPOSED NORTH STAR

Possible-risk outcomes per 100 sessions (current value)

A proxy: to be replaced by consultations booked per 100 sessions once bookings are tracked

Journey

Start and completion rates; drop-off by question and device

Acquisition

Cost per possible-risk outcome, by channel, new vs returning, age-adjusted

Data quality

Alerts on volume drops and tracking gaps

Seasonality

Year-on-year comparison, e.g. World Diabetes Day (14 Nov)

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Thank you

Questions & discussion